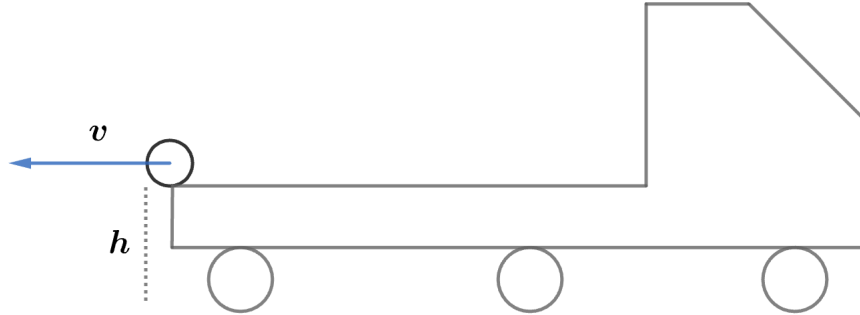


2016 F=ma Exam: Problem 16

Kevin S. Huang



We go to the truck frame where it is at rest and the ball is moving at $v = 8 \text{ m/s}$ to the left. To drop by distance h , it takes time

$$h = \frac{1}{2}gt^2$$

$$t = \sqrt{\frac{2h}{g}}$$

In this time, the ball travels a distance

$$d = vt = v\sqrt{\frac{2h}{g}} = (8 \text{ m/s})\sqrt{\frac{2(1.25 \text{ m})}{(10 \text{ m/s}^2)}} = 4 \text{ m}$$

to the left so the answer is .